

1 Casino Gambling MDP with Two Slot Machines

You are in a casino. You start with \$10 and will play until you lose all your money or reach \$30. You may choose between two slot machines: 1) Slot machine A costs \$10 to play and returns \$20 with probability 0.1 and \$0 otherwise. 2) Slot machine B costs \$20 to play and returns \$30 with probability 0.4 and \$0 otherwise. At each turn, you choose either machine A or B.

(a)

Compute the expected reward from playing machine A and machine B once.

$$A: E(r_a) = 0.1 * (20 - 10) + 0.9 * (0 - 10) = -8$$

$$B: E(r_b) = 0.4 * (30 - 20) + 0.6 * (0 - 20) = -8$$

(b)

Model this problem as an MDP. Let the state be the current amount of money. Let the action be playing either machine A or B once, and the reward be the money gained from that play. Write down the state space and action space. Draw a diagram for this MDP (similar to Example 3.3, p.52, Sutton & Barto). Use different node markings for terminal and non-terminal states.

$S=[0,10,20,30]$; $A=[A,B]$; Terminal State= $[0,30]$

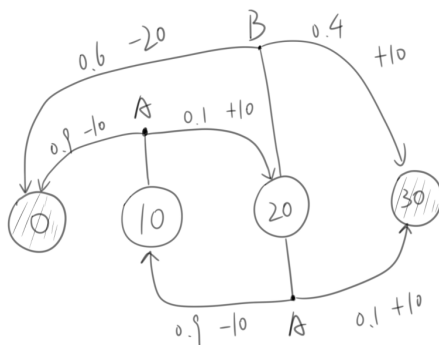
Then we can get $p(s', r|s, a)$

$$p(20, +10|10, A) = 0.1; \quad p(0, -10|10, A) = 0.9$$

$$p(30, +10|20, A) = 0.1; \quad p(10, -10|20, A) = 0.9$$

$$p(30, +10|20, B) = 0.4; \quad p(0, -20|20, B) = 0.6$$

The plot is as following:



(c)

Explain why all possible policies $\pi(a | s)$ can be uniquely defined by a parameter $\beta \in [0, 1]$, where β is the probability of choosing slot machine A when you have \$20.

Consider such a policy π_β . Compute $v^{\pi_\beta}(s)$ for all non-terminal states. What is the optimal policy?

First we know $v^{\pi_\beta}(30) = v^{\pi_\beta}(0) = 0$, According to Bellman equation,
$$v^{\pi_\beta}(20) = \beta[0.1*(10+0)+0.9*(-10+v(10))]+(1-\beta)[0.4*(10+0)+0.6*(-20+0)]$$
$$= 0.9\beta v^{\pi_\beta}(10) - 8$$

$$v^{\pi_\beta}(10) = 0.1*(10+v(20)) + 0.9*(-10+0) = -8 + 0.1v^{\pi_\beta}(20)$$

Then we can solve these two functions, get

$$v^{\pi_\beta}(10) = \frac{-8.8}{1-0.09\beta}$$

$$v^{\pi_\beta}(20) = -8 + 0.9\beta \frac{-8.8}{1-0.09\beta} = \frac{-8-7.2\beta}{1-0.09\beta}$$

To consider optimal policy, we can rewrite $v^{\pi_\beta}(20) = 80 - \frac{88}{1-0.09\beta}$

$$\text{And } v^{\pi_\beta}(10) = -\frac{8.8}{1-0.09\beta}$$

So with β increases, Both value function decreases, so the optimal policy is $\beta = 0$.

Which is, always choose B when you have \$20.

(d)

Now suppose slot machine B costs \$20 and returns \$30 with probability $0 < \eta < 1$ and \$0 otherwise. What value of η ensures that any policy is optimal?

Now $p(30, +10|20, B) = \eta$; $p(0, -20|20, B) = 1 - \eta$

Any policy is optimal means $q(20, A) = q(20, B)$

$$q(20, A) = 0.1(10 + 0) + 0.9(-10 + v(10)) = -8 + 0.9v(10)$$

$$q(20, B) = \eta(10 + 0) + (1 - \eta)(-20 + 0) = 30\eta - 20$$

$$v(10) = 0.1(10 + v(20)) + 0.9(-10 + 0) = -8 + 0.1v(20)$$

Because $q(20, A) = q(20, B)$, so $v(20) = \beta q(20, A) + (1 - \beta)q(20, B) = q(20, A)$

So $v(10) = -8 + 0.1(-8 + 0.9v(10))$

$$\text{get } v(10) = \frac{-8.8}{0.91} = \frac{-880}{91}$$

Then let $q(20, A) = q(20, B)$

$$-8 + 0.9 \frac{-880}{91} = 30\eta - 20$$

$$\text{get } \eta = \frac{10}{91} \approx 10.99\%$$

2 Sutton & Barto Exercise 4.3 and Contraction of the Action-Value Operator

(a)

Answer Sutton & Barto Exercise 4.3. Begin with

$$q^\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a],$$

and show all derivations.

$$\begin{aligned} q^\pi(s, a) &= \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a] \\ &= \mathbb{E}_\pi[R_{t+1} + \gamma G_{t+1} \mid S_t = s, A_t = a] \\ &= \mathbb{E}_\pi[R_{t+1} + \gamma v^\pi(S_{t+1}) \mid S_t = s, A_t = a] \\ &= \sum_{s', r} p(s', r \mid s, a)[r + \gamma v^\pi(s')] \end{aligned}$$

$$\begin{aligned} q^{k+1}(s) &= \mathbb{E}_\pi[R_{t+1} + \gamma v^k(S_{t+1}) \mid S_t = s, A_t = a] \\ &= \sum_{s', r} p(s', r \mid s, a)[r + \gamma v^k(s')] \end{aligned}$$

(b)

Equation (4.5) in Sutton & Barto can be written as $v_{k+1} = T_\pi(v_k)$, where T_π is an operator acting on the value function.

Analogously, define an iteration

$$q_{k+1} = T_\pi^q(q_k)$$

to compute the action-value function for policy π , where T_π^q acts on q .

Show that T_π^q is γ -contracting with respect to $\|\cdot\|_\infty$, where $0 < \gamma < 1$.

(Hint: Any q can be represented as an nm -dimensional vector, where n is the number of states and m is the number of actions. Then

$$T_\pi^q(q) = R_\pi^q + P_\pi^q q$$

for some $R_\pi^q \in \mathbb{R}^{nm}$ and $P_\pi^q \in \mathbb{R}^{nm \times nm}$. Identify these objects and determine which property of P_π^q ensures contraction.)

We need to prove that $\|T_\pi^q(q) - T_\pi^q(q')\|_\infty \leq \gamma\|q - q'\|_\infty$

$$T_\pi^q(q) - T_\pi^q(q') = R_\pi^q + \gamma P_\pi^q q - R_\pi^q - \gamma P_\pi^q q' = \gamma P_\pi^q (q - q')$$

Here P_π^q is a state-action transition probability matrix, it means the probability from one state-action position to another.

So $P_\pi^q[(s', a') \mid (s, a)] = \sum p(s' \mid (s, a))\pi(a' \mid s')$

$\sum_{s', a'} P_\pi^q[(s', a') \mid (s, a)] = 1$ because it will conclude all one possible transition ways.

So $P_\pi^q[(s', a') \mid (s, a)] \leq 1$

Then $T_\pi^q(q) - T_\pi^q(q') = \gamma P_\pi^q (q - q') \leq \gamma(q - q')$

So $\|T_\pi^q(q) - T_\pi^q(q')\|_\infty \leq \gamma \|q - q'\|_\infty$

We proved that T_π^q is γ -contracting with respect to $\|\cdot\|_\infty$.

3 Single-Server Queue Control (Bertsekas RL, 4.26)

Consider a single-server queueing system with L customers waiting. At each time step, you may choose to serve μ customers, where $\mu \in \{0, 1, \dots, L\}$. After serving μ customers, there is a cost $h(\mu)$ and an additional cost $c(i)$ for i customers remaining in the queue.

(a)

Formulate this as an infinite-horizon DP problem with discount factor $0 < \gamma < 1$. Specify the state space and action space.

State space is the number of remaining customers. $S_t \sim \{0, 1, \dots, L\}$

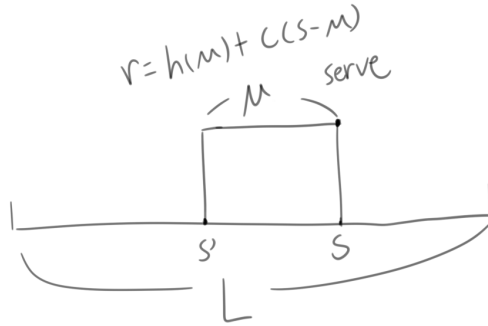
Action space is the number of served customers at one time step $A_t = \mu \sim \{0, 1, \dots, s\}$ (smaller than the current customers)

(b)

Draw the transition diagram and specify the state-reward distribution

$$p(s', r \mid s, a) = P(S_{k+1} = s', R_{k+1} = r \mid S_k = s, A_k = a).$$

The plot is as following:



$$p(s', r \mid s, a) = p(s - \mu, -h(\mu) - c(s - \mu) \mid s, \mu)$$

(c)

Assume $\mu \in \{0, 1\}$, $c(i) = ci$, and $h(0) = 0$. Compute the value functions $v^{\pi_j}(s)$ for all states and for the policies:

- i. π_1 : always serve when customers are present ($\mu = 1$), and do not serve when the queue is empty.
- ii. π_2 : always refuse to serve ($\mu = 0$).

policy 1: $v^{\pi_1}(0) = 0$ We can get the Bellman equation:

$$v^{\pi_1}(s) = -h(1) - c(s-1) + \gamma v^{\pi_1}(s-1) \text{ when } 1 \leq s \leq L$$

$$\text{So } v^{\pi_1}(s) = \sum_{k=0}^{s-1} \gamma^k [-h(1) - c(s-k-1)]$$

policy 2: $v^{\pi_2}(0) = 0$ Also can write the Bellman equation:

$$v^{\pi_2}(s) = -h(0) - c(s-0) + \gamma v^{\pi_2}(s) = -cs + \gamma v^{\pi_2}(s)$$

$$\text{So } v^{\pi_2}(s) = \frac{-cs}{1-\gamma} \text{ for } 1 \leq s \leq L$$

(d)

Show that if

$$\frac{c}{1-\gamma} > h(1),$$

then policy π_1 dominates π_2 , i.e.,

$$v^{\pi_1}(s) \geq v^{\pi_2}(s), \quad \forall s \in S.$$

We can prove it by induction, $v^{\pi_1}(0) = v^{\pi_2}(0) = 0$

$$\text{When } s=1 \quad v^{\pi_1}(s) = -h(1) - c(1-1) = -h(1)$$

$$v^{\pi_2}(s) = -\frac{c}{1-\gamma}$$

$$\therefore \frac{c}{1-\gamma} > h(1), \therefore v^{\pi_1}(s) > v^{\pi_2}(s)$$

Assuming it is correct when $s=t$, $v^{\pi_1}(t) \geq v^{\pi_2}(t)$

Then when $s=t+1$

$$v^{\pi_1}(t+1) = -h(1) - ct + \gamma v^{\pi_1}(t)$$

$$v^{\pi_2}(t+1) = \frac{-ct-c}{1-\gamma}$$

$$\begin{aligned} v^{\pi_1}(t+1) &= -h(1) - ct + \gamma v^{\pi_1}(t) \geq -h(1) - ct + \gamma v^{\pi_2}(t) = -h(1) - ct + \gamma \frac{-ct}{1-\gamma} \\ &= -h(1) - \frac{ct}{1-\gamma} = -h(1) + \frac{-ct-c}{1-\gamma} + \frac{c}{1-\gamma} = \frac{c}{1-\gamma} - h(1) + v^{\pi_2}(t+1) \end{aligned}$$

$$\text{It is } v^{\pi_1}(t+1) - v^{\pi_2}(t+1) \geq \frac{c}{1-\gamma} - h(1)$$

$$\therefore \frac{c}{1-\gamma} > h(1), \therefore v^{\pi_1}(t+1) - v^{\pi_2}(t+1) \geq \frac{c}{1-\gamma} - h(1) > 0$$

Then we get:

$$v^{\pi_1}(t+1) \geq v^{\pi_2}(t+1)$$

Induction completed, so we proved policy π_1 dominates when $\frac{c}{1-\gamma} > h(1)$

4 Equivalency Between a Discounted Problem and One with a Geometric Horizon

Consider an undiscounted MDP M with action space $\mathcal{A}$ and state space $\mathcal{S} \cup \{z\}$, where z is an absorbing terminal state:

$$P(z \mid z, a) = 1, \quad \forall a \in \mathcal{A},$$

$$r(z, a) = 0, \quad \forall a \in \mathcal{A}.$$

Transitions are time-independent. Furthermore, assume that

$$P(z \mid s, a) = 1 - \gamma, \quad \forall (s, a) \in \mathcal{S} \times \mathcal{A}.$$

The objective is to maximize the cumulative undiscounted reward:

$$\mathbb{E}_\pi \left[\sum_{t=0}^{\infty} R_{t+1} \mid S_0 = s \right], \quad s \in \mathcal{S}.$$

(a)

Consider starting at $t = 0$ at state s . Let T be the time until transitioning to the absorbing state z . Show that T is a geometrically distributed random variable. What is its parameter?

For each time step, the probability arriving terminal state z : $P(z \mid s, a) = 1 - \gamma$. So transit to the state z at time T means do not arrive z in the former $T-1$ times and arrive z in time step T .

$$\therefore P(T = t) = \gamma^t (1 - \gamma)$$

It follows geometric distribution $T \sim \text{Geo}(1 - \gamma)$ and we can calculate its parameter:

$$E[T] = \frac{1}{1 - \gamma}, \quad \text{Var}[T] = \frac{\gamma}{(1 - \gamma)^2}$$

(b)

Define the Bellman operators:

$$(T_\pi v)(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) \left(r(s, a) + \sum_{s' \in \mathcal{S} \cup \{z\}} P(s' \mid s, a) v(s') \right),$$

$$(T^* v)(s) = \max_{a \in \mathcal{A}} \left(r(s, a) + \sum_{s' \in \mathcal{S} \cup \{z\}} P(s' \mid s, a) v(s') \right).$$

Show that these operators are equivalent to those of a discounted MDP $\tilde{M}$ with state space $\mathcal{S}$ (no terminal state) and transition probabilities:

$$\tilde{P}(s' | s, a) = \frac{1}{\gamma} P(s' | s, a).$$

Since no terminal state z ,

$$\sum_{s'} P(s' | s, a) = 1 - P(z | s, a) = 1 - 1 + \gamma = \gamma$$

$$\because \tilde{P}(s' | s, a) = \frac{1}{\gamma} P(s' | s, a), \therefore \sum_{s'} \tilde{P}(s' | s, a) = \frac{1}{\gamma} \sum_{s'} P(s' | s, a) = 1$$

So logistically they are actually the same.

$$(T_\pi v)(s) = \sum_{a \in \mathcal{A}} \pi(a | s) \left(r(s, a) + \gamma \sum_{s' \in \mathcal{S}} \tilde{P}(s' | s, a) v(s') \right),$$

$$(T^* v)(s) = \max_{a \in \mathcal{A}} \left(r(s, a) + \gamma \sum_{s' \in \mathcal{S}} \tilde{P}(s' | s, a) v(s') \right).$$